D1 Problem and Research Question Proposal

Group 1
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1. Problem

Flight delays are a common problem that affects travelers, airlines, and airports. Delays can cause passengers to miss connecting flights, change their travel plans, and spend extra time waiting at airports. They can also make it harder for airlines and airports to manage their schedules. Our project aims to predict whether a flight departing from any airport will be delayed. By using past data from the airlines as well as other data sets, we want to help travelers understand the possibility of delays before their flights are actually delayed. Airlines and airports could also use these predictions to better prepare for possible disruptions.

2. Domain choice

We chose this domain because flight delays are common problems that we face when going back home for the holidays, traveling to see friends, or going on family trips. Flight delays can be frustrating because they affect our schedules and travel plans. Through this project, we hope to learn which factors have the biggest impact on flight delays, such as airlines, destinations, departure times, and travel dates. We also want to understand whether these factors can help us predict flight delays before they happen. Our goal is to use real flight data to build a useful prediction model that can help travelers make better decisions and prepare for possible delays.

3. Research question

What is the likelihood that your flight will be delayed based on your airline and destination?

Our project aims to predict the likelihood of a flight being delayed. We will use flight data and other datasets from 2023, 2024, and 2025 to identify patterns and factors that may affect flight delays. By analyzing these factors, we hope to estimate the probability of a flight being delayed and help people to better prepare for their trips.

4. Dataset

Airline Statistics:https://www.transtats.bts.gov/ONTIME/ 17 x 21 million (for just Departures)
Weather Statistics: https://open-meteo.com/ 11x32,880 (2023-2025 Daily)

Carrier Statistics: https://www.transtats.bts.gov/ot_delay/ot_delaycause1.asp 21 x 65,000 (all carriers and all airports)
Quality Statistics:
https://www.transportation.gov/individuals/aviation-consumer-protection/air-travel-consumer-reports 24 x 360 (all airports every month)

We will use flight data from 2023, 2024, and 2025. The Bureau of Transportation Statistics dataset contains records of airlines, airports, scheduled departure time, actual departure time, scheduled elapsed time, actual elapsed time, departure delay, wheels-off time, taxi-Out time, and cause of delay. We will also use previous weather data to be able to predict if it will have an effect on the possibility of a delay.

5. Evaluation plan

We can create a simple baseline: if the model can predict if a flight is delayed with 95% accuracy, then our model is working very well. We can test it on flights that are going out that day and see how the predicted delays compare to the flights that actually get delayed. With the idea of also including the number of minutes the plane is delayed by, we can also compare how delayed each flight was and how it compares to our predictions. If the accuracy is up to 95% with a good precision and recall score, we can trust the model for decision-making based on the results. While this isn't a set baseline at this current point, this is the level of accuracy we would like to evaluate at if possible.

6. Scope and risks

This project can apply to every flight coming from the United States and predict for every airport. Everyone involved with flights can use this project, as it can help airports prepare for potential delays, and people who are flying will be less surprised when they find out their flight has been delayed. A risk that can stem from this project is users trusting the model too much and ending up being late for a flight, or an airport making a miscalculation based on the percentage provided. If the model is also inaccurate, users will be frustrated finding out their flight is delayed for multiple hours when the model said it would only be 15-20 minutes.